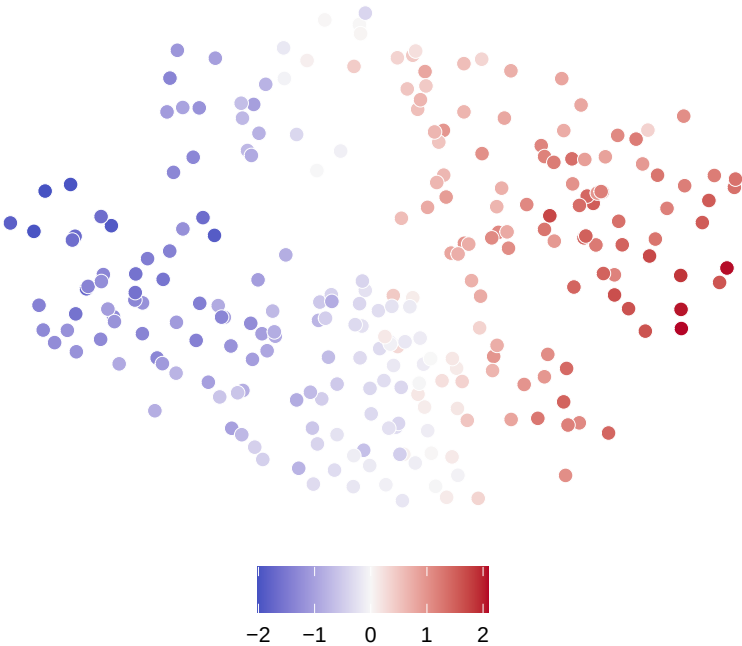


Patch initialization and one iterative refinement cycle

Patches begin as a complete spatial partition and are repeatedly reshaped, reassigned, and filtered for contiguity

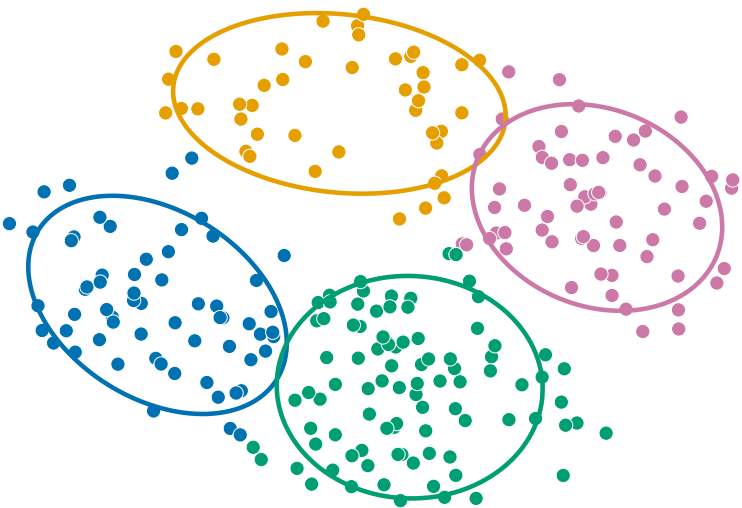
A Aligned target-cell inputs

All target cells enter together; no hotspot seeds are selected



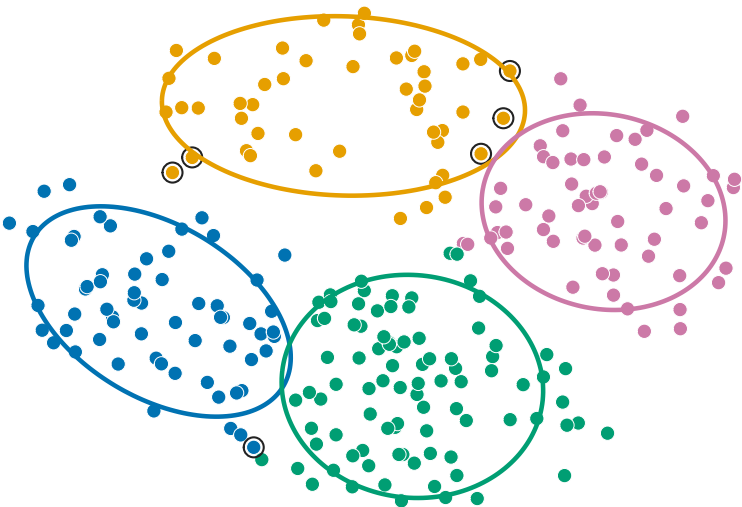
B Spatial k-means initialization

A complete starting partition defines initial centroids and ellipses



C Joint assignment and ellipse refit

Spatial, X, Z, and hunger terms can move boundary cells



D Spatial-only assignment and contiguity

Only the largest connected component of each label is retained

